Started on	Thursday, 23 June 2022, 5:11 PM
State	Finished
Completed on	Thursday, 23 June 2022, 5:20 PM
Time taken	9 mins 28 secs
Marks	7.67/9.00
Grade	**8.52** out of 10.00 (**85.19**%)

Question **1**

Correct

Mark 1.00 out of 1.00

According to R. Sutton and Minsky, what is the problem of credit-assignment?

Select one or more:

- [ ] a. Assigning credit or blame for overall episodes that contributed to the expectation of the return.
- [ ] b. Identifying actions that led to high rewards
- [x] c. Assigning credit or blame for overall outcomes to each of the decisions that contributed to those outcomes ✔
- [ ] d. Identifying models that get high rewards

Your answer is correct.

The correct answer is: Assigning credit or blame for overall outcomes to each of the decisions that contributed to those outcomes

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following sentences are true

Select one or more:

- [x] a. TD(λ) assigns credit faster than TD(0) ✔
- [ ] b. TD methods corrects the bias faster than MC
- [ ] c. Variance in TD(0) increases with the delay of the reward
- [x] d. Variance in MC increases with the delay of the reward ✔

Your answer is correct.

The correct answers are: TD(λ) assigns credit faster than TD(0)
, Variance in MC increases with the delay of the reward

Question **3**

Correct

Mark 1.00 out of 1.00

The on-site variance ω is defined as:

$\omega(s,a) = Var_r[r \mid s,a] + Var_{s',a'}[q^\pi(s',a') \mid s,a]$

In which of the following situations, this on-site variance is zero?

Select one or more:

- a. For non-delayed rewards environments
- b. For TD(0) and deterministic rewards
- c. For deterministics state transition environments
- d. For deterministics state transition environments, deterministic rewards and deterministic policies. ✔

Your answer is correct.

The correct answer is: For deterministics state transition environments, deterministic rewards and deterministic policies.

Question **4**

Correct

Mark 1.00 out of 1.00

In MC, and under certain assumptions, the number of states affected by the on-site variance is:

$a_k = \sum_{t=0}^{k} \left(1 - (1 - \frac{c_t}{N_{t-1}})^{n_t}\right) N_{t-1}$

where N_t is the number of states reachable after t steps, n_t is the number of states affected by the on-site variance at time t and c_t is the random average connectivity of s_t to s_{t-1}.

What of the following sentences concerning k can be deduced from the expression above? (collorary of the theorem)

Select one or more:

- a. For **large** k, and after some time $t > \hat{t}$, the number of states affected by the on-site variance at step k **grows linearly** with k ✔
- b. For **large** k, the number of states affected by the on-site variance at step k **grows exponentially** with k
- c. For **small** k, the number of states affected by the on-site variance at step k **grows exponentially** with k ✔
- d. For **small** k, and after some time $t > \hat{t}$, the number of states affected by the on-site variance at step k **grows linearly** with k

Your answer is correct.

The correct answers are: For **small** k, the number of states affected by the on-site variance at step k **grows exponentially** with k, For **large** k, and after some time $t > \hat{t}$, the number of states affected by the on-site variance at step k **grows linearly** with k

Question **5**

Partially correct

Mark 0.67 out of 1.00

If we make the expected future rewards equals to zero (i.e. via reward redistribution), the Q-values are just the average of the immediate reward. However, why this is not possible in an MDP setting?

Select one or more:

- [ ] a. Because in this case (future expected reward equal to zero), the reward is obtained before we actually get in in the environment, and therefore, is not causal and breaks the Markov property.
- [x] b. Because in order to have future expected reward to zero, states have to be enriched with the past, and therefore, breaks the Markov assumption ✖
- [ ] c. It is completely possible in an MDP setting.
- [x] d. Because in this case (future expected reward equal to zero), the reward depends also on the previous state and action, and therefore, is not Markov. ✔

Your answer is partially correct.

You have selected too many options.
The correct answer is: Because in this case (future expected reward equal to zero), the reward depends also on the previous state and action, and therefore, is not Markov.

Question **6**

Correct

Mark 1.00 out of 1.00

What is the definition of a sequence-Markov decision process (SDP), as seen in class?

Select one or more:

- [ ] a. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov reward** and has **Markov policy** but a **transition probabilities that is not required to be Markov**
- [x] b. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** and has **Markov transition probabilities** but a **reward that is not required to be Markov** ✔
- [ ] c. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov reward** and has **Markov transition probabilities** but a **policy that is not required to be Markov**
- [ ] d. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** but a **transition probabilities and reward that is not required to be Markov**

Your answer is correct.

The correct answer is: A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** and has **Markov transition probabilities** but a **reward that is not required to be Markov**

Question **7**

Incorrect

Mark 0.00 out of 1.00

What is the dfinition of **return-equivalent** SDPs?

Select one or more:

- [ ] a. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same return per episode**
- [ ] b. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return**
- [x] c. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return per episode** ✖

Your answer is incorrect.

The correct answer is: Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return**

Question **8**

Correct

Mark 1.00 out of 1.00

What is the **explaining away problem** in MDPs, when a function g is used to predict the return based on the sequence of actions and states?

Select one or more:

- [x] a. Since later states are used for return prediction, earlier actions causing reward are missed. ✔
- [ ] b. Since the sequence of states and actions are mutually independent, the prediction needs every component to explain the return.
- [ ] c. Previous actions can explain away future states, breaking the Markov property
- [ ] d. In an MDP, one only need the difference of states actions to make a prediction and not the whole sequence.

Your answer is correct.

The correct answer is: Since later states are used for return prediction, earlier actions causing reward are missed.

Question **9**

Correct

Mark 1.00 out of 1.00

What of the following statements are true when refering to differences between LSTM and Feed Forward networks (FFN) for making future predictions?

Select one or more:

- [x] a. LSTM will only learn to store information if a state-action pair has a strong evidence for a change in the expected return. ✔
- [x] b. LSTM tends to have smaller prediction errors since reuses past information ✔
- [x] c. Prediction errors of LSTMs are much more likely to cancel via prediction differences. Since consecutive predictions of LSTMs rely on the same internal states, they usually have highly correlated errors. ✔
- [x] d. FFNs are not suited for processing sequences. ✔
- [x] e. FFNs are prone to prediction errors since they have to extract information from every state to make a prediction in every timestep ✔

Your answer is correct.

The correct answers are: FFNs are not suited for processing sequences., FFNs are prone to prediction errors since they have to extract information from every state to make a prediction in every timestep, LSTM will only learn to store information if a state-action pair has a strong evidence for a change in the expected return., LSTM tends to have smaller prediction errors since reuses past information, Prediction errors of LSTMs are much more likely to cancel via prediction differences. Since consecutive predictions of LSTMs rely on the same internal states, they usually have highly correlated errors.

◄ Direct Credit Assignment

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